Grouping and Aggregation

- Tuples with the same attribute values (for a specified list of attributes) are grouped.
- An aggregate function is applied to each group (computing one value for each group).
- Typical aggregate functions
  - count: number of tuples in a group
  - sum: sum of attribute values in a group
  - min, max, avg: minimum, maximum, average of attribute values in a group
- Notation: $\gamma_{L;\,F}(r)$
  - L list of attributes for grouping
  - F aggregate function

Grouping and Aggregation

- Example: determine the number of students per semester
- $\gamma_{semester;\ count(*)}(student)$

student

name	*semester*	*age*
Alice	7	24
Anna	3	20
Bob	3	19
Cathy	5	22
Christian	5	20
David	1	18

Result

semester	*count(*)*
7	1
3	2
5	2
1	1

Grouping and Aggregation

- Example: determine the number of students per semester and the minimum age in each group
- $\gamma_{semester;\ count(*),\ min(age)}(student)$

student

name	*semester*	*age*
Alice	7	24
Anna	3	20
Bob	3	19
Cathy	5	22
Christian	5	20
David	1	18

Result

semester	*count(*)*	*min(age)*
7	1	24
3	2	19
5	2	20
1	1	18

Equivalent Queries

- There is more than one way to write a query in relational algebra.
- Example: find instructors in the Physics department with salary greater than 90,000
- Query 1
  - $\sigma_{dept_name = \text{“Physics”} \wedge salary > 90{,}000}(instructor)$
- Query 2
  - $\sigma_{dept_name = \text{“Physics”}}(\sigma_{salary > 90.000}(instructor))$
- The two queries are not identical; they are, however, equivalent -- they give the same result on any database.

Summary

- Relational Model
  - Relation, tuple, attribute
  - Schema and instance
  - Keys: Superkey, candidate key, primary key, foreign key
- The Relational Algebra
  - select: σ
  - project: Π
  - Cartesian product: x
  - join: $\bowtie$
  - union: $\cup$
  - set-intersection: $\cap$
  - set-difference: –
  - rename: ρ